Unit of Observation
Each observation in this dataset represents a single image of a flower.
Each image is labeled with a flower category (1–102) and is associated with a predefined train, validation, or test split.

Dataset Summary
- Total observations: 8,000+ images
- Data type: Image classification dataset
- Source: Oxford 102 Flower Dataset (University of Oxford)
- Frequency: Not time-based (cross-sectional image data)
- Number of classes: 102 flower categories
- Observations per class: Between ~40 and 250 images

Data Processing (from code)
- Labels were loaded from imagelabels.mat
- Labels were converted from 1–102 → 0–101 for modeling
- A custom 80/10/10 split was created using stratified sampling:
    - Training: ~6,551 images
    - Validation: ~819 images
    - Test: ~819 images
- Splits were verified to have no overlap

Variable Descriptions

1. Image

- Type: JPEG image (RGB pixel data)
- Description: The raw image of a flower used as the input to the model.
- Notes:
    - Images are loaded using PIL and converted to RGB
    - Pixel values are scaled to the range [0, 1]
    - Images vary in:
        - size
        - lighting
        - background
        - orientation

    - Images are resized during preprocessing (e.g., 64×64 or model-specific size)

2. Label (all_labels)

- Type: Integer (0–101)
- Description: Numeric label representing the flower category for each image
- Notes:
    - Originally provided as 1–102 in the dataset
    - Converted to 0–101 for compatibility with PyTorch
    - Used as the target variable in the classification model
    - There are 102 unique classes

3. Image Index

- Type: Integer
- Description: Index corresponding to each image file (e.g., image_00001.jpg)
- Notes:
    - Used to match images with labels
    - Constructed implicitly through file naming in the dataset
    - Used when loading images from disk

4. Training Set Indicator (train_ids)

- Type: Integer index subset
- Description: Indices of images used for model training
- Notes:
    - Contains ~80% of the dataset
    - Created using stratified sampling to preserve class balance
    - Used to fit the neural network model

5. Validation Set Indicator (val_ids)

- Type: Integer index subset
- Description: Indices of images used for model validation
- Notes:
    - Contains ~10% of the dataset
    - Used for:
        - tuning hyperparameters
        - monitoring model performance during training

6. Test Set Indicator (test_ids)

- Type: Integer index subset
- Description: Indices of images used for final model evaluation
- Notes:
    - Contains ~10% of the dataset

- Not used during training
- Provides unbiased performance estimates

Derived / Processed Variables

7. Pixel Intensity Values

- Type: Float (0–1 range)
- Description: Normalized pixel values extracted from images
- Purpose:
    - Standardizes image input
    - Improves model training stability

8. Channel Distributions (RGB)

- Type: Numeric distributions
- Description:Distribution of red, green, and blue pixel intensities
- Purpose (from EDA):
    - Understand color composition of images
    - Compare to standard normalization values (ImageNet means)

9. Class Counts

- Type: Integer
- Description: Number of images in each flower category
- Purpose:
    - Assess class imbalance
    - Identify largest and smallest classes

10. Intra-Class Variability (class_std)

- Type: Float
- Description: Average pixel-level standard deviation within each class
- Purpose:
    - Measures how visually similar or different images are within a class
    - Higher values indicate more variation (harder classification)

Exploratory Plots and Figures

Figure 1: Image Count per Class

Bar plot showing the number of images for each of the 102 flower classes.
Includes:

- mean and median reference lines
- identification of largest and smallest classes

Insight:
There is noticeable class imbalance, with some classes having several times more images than others.

Figure 2: Pixel Intensity Distribution

Histogram of red, green, and blue pixel intensities across a sample of images.

Insight:

- Pixel values are distributed between 0 and 1 after normalization
- Distributions are compared to standard ImageNet normalization values

Figure 3: Sample Images by Class

Grid of randomly selected images from multiple flower classes.

Insight:

- High variability in background, lighting, and orientation
- Demonstrates complexity of classification task

Figure 4: Intra-Class Variability

Bar plot showing variation within each class using pixel standard deviation.

Insight:

- Some classes are visually consistent (easy to classify)
- Others show high variation (more difficult for the model)

Figure 5: Example Images – Most Consistent vs Most Diverse Classes

Displays images from:

- most visually consistent classes
- most visually diverse classes

Insight:

- Highlights differences in classification difficulty across categories